

Consider the customer-sale scenario given below. The primary keys are underlined and the data types are specified:

CUSTOMER(Cust id : integer, cust_name: string)

ITEM(item_id: integer, item_name: string, price: integer)

SALE(bill_no: integer, bill_data: date, cust_id: integer, item_id: integer, qty_sold: integer)

For the above schema, perform the following:

- Create the tables with the appropriate integrity constraints
- Insert around 10 records in each of the tables
- List all the bills for the current date with the customer names and item numbers
- List the total Bill details with the quantity sold, price of the item and the final amount
- List the details of the customer who have bought a product which has a price>200
- Give a count of how many products have been bought by each customer
- Give a list of products bought by a customer having cust_id as 5
- List the item details which are sold as of today
- Create a view which lists out the bill_no, bill_date, cust_id, item_id, price, qty_sold, amount

```
mysql> create database customer_sale;
```

```
mysql> use customer_sale;
```

a. Create the tables with the appropriate integrity constraints

```
mysql>create table customer(custid integer(5) primary key, custname  
varchar(25));
```

```
mysql>create table item(item_id integer(5) primary key, item_name  
varchar(20),price integer(5));
```

```
mysql>create table sale(bill_no integer(5) primary key,bill_date date,  
cust_id integer(5) references customer(custid),itemid integer(5)  
references item(item_id),qty_sold integer(5));
```

```
mysql> show tables;
```

```
+-----+  
| Tables_in_customer_sale |  
+-----+  
| customer                |  
| item                    |  
| sale                    |  
+-----+
```

3 rows in set (0.06 sec)

```
mysql> desc customer;
```

```
+-----+-----+-----+-----+-----+-----+  
| Field | Type   | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| custid | int    | NO   | PRI | NULL    |      |  
| custname | varchar(25) | YES |     | NULL    |      |  
+-----+-----+-----+-----+-----+-----+
```

2 rows in set (0.08 sec)

b) Insert around 10 records in each of the tables.

mysql> insert into customer values(1,'custA');
Query OK, 1 row affected (0.01 sec)

mysql> insert into customer values(2,'custB');
Query OK, 1 row affected (0.07 sec)

mysql> insert into customer values(3,'custC');
Query OK, 1 row affected (0.07 sec)

mysql> insert into item values(101,'itemA',100);
Query OK, 1 row affected (0.06 sec)

mysql> insert into item values(102,'itemB',1000);
Query OK, 1 row affected (0.01 sec)

mysql> insert into item values(103,'itemC',5000);
Query OK, 1 row affected (0.07 sec)

mysql> insert into sale values(1001,'2024-01-05',1,101,15);
Query OK, 1 row affected (0.07 sec)

mysql> insert into sale values(1002,'2024-01-01',2,102,5);
Query OK, 1 row affected (0.07 sec)

mysql> insert into sale values(1003,'2024-01-01',2,101,15);
Query OK, 1 row affected (0.01 sec)

mysql> select * from customer;

custid	custname
1	custA
2	custB
3	custC

3 rows in set (0.00 sec)

mysql> select * from item;

item_id	item_name	price
101	itemA	100
102	itemB	1000
103	itemC	5000

3 rows in set (0.00 sec)

```
mysql> select * from sale;
```

bill_no	bill_date	cust_id	itemid	qty_sold
1001	2024-01-05	1	101	15
1002	2024-01-01	2	102	5
1003	2024-01-01	2	101	15
1004	2025-03-26	1	102	15

4 rows in set (0.00 sec)

- c) List all the bills for the current date with the customer names and item numbers

```
mysql> select custname,itemid from sale,customer where  
sale.cust_id=customer.custid and sale.bill_date=curdate();
```

custname	itemid
custA	102

1 row in set (0.00 sec)

- d) List the total Bill details with the quantity sold, price of the item and the final amount.

```
mysql> select itemid,item.price,sum(qty_sold) from sale,item where  
sale.itemid=item.item_id group by sale.itemid, price;
```

*(Note: the below query will also work if your version compatible with
sql_mode=only_full_group_by (without price in group by column))*

```
mysql> select itemid,item.price,sum(qty_sold) from sale,item where  
sale.itemid=item.item_id group by sale.itemid;  
)
```

itemid	price	sum(qty_sold)
101	100	30
102	1000	20

2 rows in set (0.00 sec)

- e. List the details of the customer who have bought a product which has a price>200

```
mysql> select custid,custname,itemid,price from customer,item,sale
where sale.cust_id=customer.custid and sale.itemid=item.item_id and
item.price>200;
```

custid	custname	itemid	price
2	custB	102	1000
1	custA	102	1000

2 rows in set (0.07 sec)

- f. Give a count of how many products have been bought by each customer.

```
mysql> select cust_id,count(*) from sale group by cust_id;
```

cust_id	count(*)
1	2
2	2

2 rows in set (0.00 sec)

- g. Give a list of products bought by a customer having cust_id as 5.

```
mysql> select itemid as items_bought_by_2 from sale,item where
sale.itemid=item.item_id and cust_id=2;
```

items_bought_by_2
101
102

2 rows in set (0.00 sec)

- h. List the item details which are sold as of today.

```
mysql> select curdate(),itemid from sale where bill_date=curdate();
```

curdate()	itemid
2025-03-26	102

1 row in set (0.01 sec)

- i. Create a view which lists out the bill_no, bill_date, cust_id, item_id, price, qty_sold, amount.

```
mysql> create view view_sale_info as(select
cust_id,custname,itemid,item_name,bill_no,bill_date,qty_sold, price
from customer,sale,item where sale.cust_id=customer.custid and
sale.itemid=item.item_id);
```

Query OK, 0 rows affected (0.09 sec)

```
mysql> show tables;
```

```
+-----+
| Tables_in_customer_sale |
+-----+
| customer                |
| item                    |
| sale                    |
| view_sale_info          |
+-----+
```

4 rows in set (0.01 sec)